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Perception of urban soundscape and landscape using different visual environment reproduction methods in virtual reality

Hyun In Jo [1], Jin Yong Jeon *

Department of Architectural Engineering, Hanyang University, Seoul 04763, South Korea

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ABSTRACT

This study investigated the effects of visual environment reproduction methods of virtual reality (VR) on soundscape and landscape assessments for reproducing VR environments in laboratory settings. It also subjectively assessed urban environments to compare assessment method qualities. Specific monitor and head-mounted display (HMD) environments were selected as visual reproduction methods. Spatial audio that applied first-order ambisonic-based head-tracking technology was equally provided in both visual environments. We analyzed the subjective responses of 40 individuals according to overall satisfaction, VR reproduction quality, perceived audio-visual elements, and adjective attributes describing soundscape and landscape. The results showed that urban environments were more sensitively assessed through HMD than the monitors. Negative and positive components were more clearly perceived in the HMD environment. Furthermore, the negative effects of artificial components were rated highly in the HMD environment. The surrounded landscape affected urban contextual perceptions positively in HMD and negatively in monitor environments, making it an important assessment factor. Consequently, monitor and HMD environments were beneficial to the overall awareness of urban environments and spatial presence, respectively. The discoveries and data of this study could be extensively used as a basis for developing experimental methodologies to assess the soundscape and landscape of urban environments in laboratory settings.

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1. Introduction

Under the limited conditions of laboratory settings, evaluations of human psychological responses to various real spaces could not be free from ecological validity issues, despite the benefit of controlling uncontrolled and extraneous factors that occur in real-life settings [1–4]. In this context, virtual reality (VR) technology has secured a sense of presence with high ecological validity [5], playing a progressive role in assessing and predicting real-world human behavioral responses in laboratory settings [6,7]. Initially, education on and assessments of cognitive responses were mostly implemented in indoor environments [8,9], but they were eventually extended to the outdoors, going as far as stakeholders (architects, urban planners, and policymakers) attempting to share information about actual spatial planning [1,10]. As VR technology has become widespread, studies have actively evaluated its utility and applicability in urban environments, and many have verified the utility of environmental preferences or cognitive assessments in VR environments [11–14].

Multisensory design is very important for considering holistic perceptions when analyzing urban environments. Many studies have examined the effects of audio-visual interactions on urban environment perceptions [15–19]. Drettakis et al. [20] enabled users to regulate audio-visual factors within VR environments, such as 3D sounds, high-detail vegetation, and shows in urban contexts, all of which are useful in actual urban design and development. Ruotolo et al. [21] predicted assessment results for undeveloped spaces by examining the detrimental audio-visual effects of motorway designs in VR environments. Sanchez et al. [22] implemented dynamic 3D auralization technology to examine the effects of noises on the overall appreciation of urban public spaces in VR environments, presenting visual design guidelines for noise barriers. Maffei et al. [23] discovered that congruency between VR and real-world environments was sufficient to perceive acoustic and visual stimuli, implying that VR technology could be used to examine audio-visual interactions in urban

* Corresponding author at: Architectural Acoustics Lab (Room 605-1), Department of Architectural Engineering, Hanyang University, 222 Wangsimni-ro, Seongdong-gu, Seoul 04763, South Korea.

E-mail addresses: best2012@hanyang.ac.kr (H.I. Jo), jyjeon@hanyang.ac.kr (J.Y. Jeon).

[1] Address: Architectural Acoustics Lab (Room 605-1), Department of Architectural Engineering, Hanyang University, 222 Wangsimni-ro, Seongdong-gu, Seoul 04763, South Korea.

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environment perceptions. Based on previous studies [15–23], the head-mounted display (HMD) was the most used visual reproduction method for VR environments.

Comparative studies have examined the fidelity of sound and visual environment reproduction methods to assess monaural, binaural, and ambisonic sound methods in VR-based urban environments [24,25]. Others have examined various assessment setups to compare performances of 2D- and 3D-array sounds in speaker environments, or monaural and binaural sound performances in headphone environments [25–29]. Among these studies, Hong et al. [25] determined that the first-order ambisonic (FOA) binaural setup reflecting head movements sufficiently secured the urban environment's acoustic fidelity, and many studies conducted since have applied this sound-reproduction method [17,19,29]. However, minimal studies have assessed the effects of different visual elements on urban environment perceptions. Considering that the spatial perception gaps are considerable within the visual reproduction methods in computer science and cognitive behavior, further research must also investigate these perceptions in urban environments [30–33]. However, if VR application to other fields is focused on immersion, emotional meaning must also be considered in human perception [34]. Although attempts have been made to compare 2D and 3D landscape images and videos, such comparisons are limited by differences in realism functionally depending on image precision [35] or by preferences of urban planning stakeholders regarding 2D and 3D images [36]. In assessing urban environments, it is very important to develop an immersive VR environment to better represent the actual urban users' emotions. Thus, it is necessary to examine the differences in perception elements according to visual reproduction methods.

Recently, various platform-based assessment tools, such as web, mobile, and HMD, have been developed to design and plan urban environments, and 3-D visualization and auralization technologies have been actively incorporated therein [37–40]. However, most tools are based on animated virtual spaces, instead of real-life, concentrating on auditory elements to secure fidelity to real spaces and selectively using visual reproduction methods only when necessary. Previous studies [26,41] have sufficiently verified the ecological validity of sound-reproduction methods and intensively covered perception differences among assessment environments. However, because urban environments are affected by audio-visual interactions, it is necessary to deeply examine the perception differences in urban contexts according to different visual reproduction methods and develop appropriate urban assessments accordingly.

Therefore, this study examined the effects of visual environment reproduction methods on soundscape and landscape perceptions in VR urban environments to determine how overall satisfaction varies, whether there were any differences in VR reproduction quality, how perception gaps appeared in audio-visual elements and the effects of each element, and how perceptions (emotive expressions) emerge in soundscapes and landscapes. Based on previous VR environment studies reviewed above, we selected a monitor (2D) and an HMD (3D) as the most representative means of projecting photos before and after the appearance of VR technology, respectively, for our visual environment reproduction methods. VR environments were implemented for each method and urban environments were subjectively assessed. Furthermore, perception gaps in each environment were analyzed and compared, and the distinctive perception results appearing only in each environment were examined. The latter ultimately presented new points to consider for selecting visual environment reproduction methods to develop VR environments in laboratory settings.

The distinctive contribution of this study is as follow. We present how perception of audio-visual environment other than immersion changes according to visual reproduction method when the VR environment is developed in laboratory settings. This shows differentiation from the previous studies in the field of soundscape that have been reviewed as above, which have mainly focused on the comparison of sound environment reproduction methods. The findings of this study are expected to be utilized for researchers who implement soundscape and landscape evaluation environments in laboratory settings with practical limitations.

2. Methods

2.1. Site selection

Eight sites in Seoul were selected to examine the effects of visual reproduction methods on soundscape and landscape assessments of urban environments in laboratory settings. Fig. 1 presents the details and functional characteristics of each site. The sites had various land uses and were selected to broaden the sound sources and visual elements. Although Sites C and F are both green spaces, Site C includes a wide rest area, whereas Site F is a walkway. Regarding public spaces, Site E usually has higher population traffic than Site H. As Site G is a rest area at a higher elevation than the other sites, it offers an expansive view.

Table 1 shows the physical properties (sound strength, spectral contents, and temporal variations) and psycho-acoustical properties of the sound environments measured at each site. In this case, dBTrait software (01 dB) and ArtemiS SUITE software (HEAD acoustics) were used for analysis. Sound strength was obtained by calculating the A-weighted sound-pressure level (SPL, L_{Aeq}) and percentile SPLs (L_{A10} and L_{A90}). Spectral contents were obtained by analyzing $L_{Ceq\text{-}Aeq}$, calculated by the difference between A and C-weighted SPLs and showing relatively low-frequency sound source features. Temporal variability was calculated by $L_{\mathrm{A10\text{-}A90}}$. The mean L_{Aeq} of the eight sites was distributed in the broad range of 57.2–79.4 dBA. For SPL, L_{Aeq} was the lowest at Site H (public space) and the highest at Site A (roadside). Moreover, $L_{A10\text{-}A90}$ and $L_{Ceq\text{-}Aeq}$ were distributed in ranges between 5.6–15.8 and 6.6–16.5 dB, respectively, indicating that the maximum and minimum values surpassed 10 dB, and were thus sufficiently within the usual variation range of sounds in urban environments. Loudness also showed a similar trend to the sound pressure level, and the value was highest at 7.27 sone at Site A close to the roadside and lowest at 0.87 sone at Site H surrounded by buildings. When examining other psychoacoustic parameters, Sharpness showed the opposite trend to $L_{\mathrm{Ceq\text{-}Aeq}}$ and the value ranged from 1.23 to 1.76 acum, Fluctuation strength ranged from 0.00 to 0.01 vacil, Roughness was in the range of 0.07 – 1.54 asper, and Tonality was 0.2 tu or less in all sites. As a result, there were no assessment sites showing outliers in the results of psychoacoustic indicators. Considering all acoustical parameters, it can be regarded that the assessment points used in this study sufficiently cover the general soundscape of the urban environment.

2.2. Stimuli

2.2.1. Audio-visual recordings

Audio-visual data were measured at the eight sites during daytime (10:00 am–2:00 pm) in September 2018, when the temperature range was 21.6–22.8 °C, and the weather was clear. Recorded in an A-format first-order ambisonic setup, the auditory data were obtained using a 4-channel ambisonic microphone (Soundfield SPS200, Soundfield Ltd.) and a portable sound recorder (MixPre-6, Sound Devices). For sound-pressure calibration, a 1.27-cm (1/2-inch) microphone (GRAS 46AE, GRAS Sound & Vibration) and a portable sound-level meter (AS-70, RION, Co., Ltd.) were used

Fig. 1. Views from eight sites and corresponding detailed descriptions.

Table 1
Acoustic and psycho-acoustic parameters in 8 different sites.

Site	A	B	C	D	E	F	G	H
L_{Aeq} [dB]	79.40	71.10	68.35	69.40	65.70	72.75	65.10	57.15
$L_{A10\text{-}A90}$ [dB]	12.50	12.30	10.10	8.90	11.55	15.75	5.65	11.40
$L_{Ceq\text{-}Aeq}$ [dB]	16.50	9.50	13.40	16.45	6.60	9.20	7.70	9.35
Loudness [sone]	7.27	3.73	2.86	3.51	2.09	3.46	2.42	0.87
Sharpness [acum]	1.76	1.66	1.64	1.58	1.35	1.34	1.33	1.23
Fluctuation strength [vacil]	0.01	0.00	0.00	0.00	0.01	0.01	0.00	0.00
Roughness [asper]	1.54	0.84	0.56	0.75	0.35	0.56	0.49	0.07
Tonality [tu]	0.05	0.04	0.09	0.06	0.10	0.18	0.09	0.03

to measure L_{Aeq}. The sampling rate was 24-bit 44,100 Hz. While recording auditory data, we simultaneously recorded the 3D visual environment data. The microphone and camera data were collected at the same position. We used a 6-channel, 360° camera (Insta360™ Pro, Insta360®) and recorded omnidirectional videos at 95 Mbps in 8 k-ultrahigh definition and at 30-fps resolution. Both visual and auditory data were measured for 3 min, and camera and microphone-tripod heights were adjusted to 1.6 m, according to ISO 12913-2 [24] recommendations. To adjust video and audio synchronization in reproduction, clapping was chosen to signal the beginning of measurements.

2.2.2. VR environment reproduction

We used open-type headphones (HD-650, Sennheiser) to provide auditory data in laboratory settings and an audio interface (AudioBox 1818VSL, PreSonus®) for desktop connection. We employed a spatial audio application (resonance audio) programming interface (API, Google VR), embedded in cross-platform game-engine software (Unity Technologies, Inc.), to implement stereophonic sounds. Instead of using the sound collected by the microphone of the VR camera, which causes interference due to the noise of the hardware, we used a separate ambisonic microphone. Based on previous research [25], we converted the sound sources recorded in A-format FOA to B-format FOA with MixPre-6 Plugin (Sounddevices); then, the sound sources were downmixed in a binaural track. For sound-pressure calibration, a head-and-torso simulator (Type 4100, Brüel & Kjær) and a NEXUS charge amplifier (Type 2690-A, Brüel & Kjær) were used to record the stereophonic sound played on headphones. SPL was then adjusted using Adobe Audition (version 1.5, Adobe) to ensure that the sounds were equivalent to the L_{Aeq} of the sound sources recorded with a calibration microphone in the actual settings.

The videos were recorded on six channels and edited by post-processing with 4 k resolution (Insta360® STITCHER, Insta360™). The completed videos were played identically in the monitor and HMD environments. As shown in Fig. 2(a), the monitor was an 80-cm (31.5-inch) (16:9 aspect ratio) flat screen with a quad high-definition (QHD) resolution of 2,560 × 1,440 pixels and a 178° angle of view (AOC Q3279, Alphascan Display). The monitor showed excellent performance with a maximum refresh rate of 75 Hz, optical angle of view (176(H)/176(V)), and brightness of 250 cd/m^2. The HMD environment had a 2,880 × 1,600-pixel resolution and a dual AMOLED 8.9-cm (3.5-inch) screen (VIVE Pro, HTC) with a 90-Hz refresh rate, 110° angle of view, and brightness of 143 cd/m^2 (Fig. 2(b)). For both visual environment reproduction methods, we used the same desktop and high-quality graphic cards (GeForce® GTX 1080 Ti, NVIDIA), processors (Intel® Core™ i5-8600K CPU @ 3.60 GHz, Intel®), and 16.0-GB memory cards (Samsung). The equipment and software used in this study does not represent the performance of all monitors and HMDs. For

Fig. 2. Subjective evaluation setups for (a) monitor and (b) HMD environments.

example, different HMD hardware may affect the angle of view, which in turn may affect immersion. Similarly, the use of stereophonic software other than Google SDK may produce different results. However, in this study, we investigated whether there is a difference in urban environment perception according to the 2D and 3D visual environments, so one type was selected as the representative equipment.

The monitor environment was constructed to allow the participants to see any direction using the mouse interface, while the HMD environment was constructed to permit participants to change the view simply by rotating their heads. Stereophonic sounds were provided through the same headphones in both visual environments. Although real-time directivity was equally implemented according to the head movements trackers of the VR device and the changing views of the mouse interface, it could be implemented only in fixed locations without considering physical movement other than that of the head. We installed the monitor ensuring that its center was located at the individual's head position at a 1-m distance. As a result, the sounds in both environments were identical, with the visual environment reproduction method being the only difference.

2.3. Subjective assessments

2.3.1. Questionnaire

We developed a questionnaire using various perceptual indicators and descriptors to determine changes in soundscape and landscape perceptions according to visual reproduction methods. Descriptors assess how humans perceive audio-visual environments and how indicators predict descriptors [42]. The survey contained questions on overall satisfaction (O), VR reproduction quality (R), perceived audio-visual elements (I), soundscape adjective attributes (S), and landscape adjective attributes (L) (Appendix A). Only the VR environment responses were rated on a 7-point Likert scale [43,44], while the others used a 5-point Likert scale [24].

The survey questions covered the following items (Appendix A). Overall satisfaction contained descriptors that are considered important in determining soundscape and landscape preferences in urban environments: overall impression [16,35], acoustic satisfaction [45,46], visual aesthetic [47,48], and comfort [49,50]. As VR environments developed in laboratory settings cannot equally implement all the stimuli present in actual environments, we posed realism (the degree of being realistic relative to the real world) and immersion (presence and envelopment) questions to assess the reproduction quality [43,51]. We divided the perceived audio-visual elements into sound and visual factors for assessment. Sound elements are important because they provide significant information on soundscape quality analysis [52]. Brown et al. [49] classified urban environment sound sources into human and facility sounds and other biophonic or geophonic sounds. Based on previous holistic soundscape studies, Axelsson et al. [53] proposed five sound classifications: traffic, fans, other noises, human sounds, and natural sounds [54,55]. In this study, traffic, human sounds (speech and other activities), birdsongs, music, and machines, commonly used in recent soundscape literature [17,56,57], were the selected sound classifications. The visual elements, most of which are categorized by natural elements or land use, provide important information for designing landscapes [58–60]. Referring to the categories previously used in the literature to objectively assess landscapes [17,37,61,62], this study selected vehicles, roads, sidewalks, high buildings, low buildings, open spaces, people, vegetation, and the sky. Many studies investigated soundscape perception through verbal descriptions or semantic differential attributes based on which various soundscape factors were derived through principal component analysis (PCA) [63–66]. Particularly, Axelsson et al. [67] proposed a 2D model (pleasantness and eventfulness) that has been used most frequently in subsequent research to analyze soundscapes. Accordingly, this study selected eight adjectives: pleasant, unpleasant, eventful, uneventful, exciting, monotonous, calm, and chaotic. Based on many landscape studies that have used semantic expressions to analyze human perception [68–71], this study selected the following 11 adjectives [17,72–75]: attractive, comfortable, uninteresting, harmonious, complex, open, repulsive, interesting, enclosed, simple, and disharmonious.

2.3.2. Procedure

We selected 40 (30 male and 10 female) individuals to participate in the study. All provided informed consent in writing. Twenty participants were randomly assigned to monitor (16 men and 4 female) and HMD (14 men and 6 female) assessment environments, developed according to visual environment reproduction methods, thereby meeting the number of required participants suggested by ISO 12913-2 [24]. To reduce response variations in the groups, only students who attended the same university and who were familiar with the assessment sites were included. Their ages ranged from 21 to 30 (Monitor: mean age = 24.90, standard deviation = 1.83; HMD: mean age = 24.40, standard deviation = 2.01), and their audibility limits were assessed using an audiometer (AA-77, RION Co., Ltd.), revealing that all had normal-level hearing. We provided simple training on experimental theory and purpose to ensure that the participants sufficiently understood the survey items before the experiment [76]. As not all the participants were familiar with using a VR device, a brief training session was carried out to help them adjust to the assessment setup.

We provided the sound source assessments in the A to H order, based on the common routes traveled by the participants. The sounds were played for 3 min in 10-s intervals, and, if the participants desired, the audio-visual stimuli were repeatedly provided at each site. To mitigate fatigue due to the long experiment, the total experiment time did not exceed 30 min. The participants responded to all the survey items at each assessment point, resulting in eight sets of responses from each subject and a total of 320 collected data sets (two visual environments × 20 participants × 8 sites). All the experiments were performed in a semi-anechoic chamber, wherein the background noise was very low (~25 dBA).

2.4. Statistical analysis

We analyzed the data using the Statistical Package for the Social Sciences (SPSS® version 25, IBM) software. First, we conducted the normality (Shapiro–Wilk and Kolmogorov–Smirnov tests) and homoscedasticity (Levene's test) tests on the two group's responses. As normality was not achieved, the nonparametric Mann–Whitney *U* test was performed to compare mean responses. The soundscape and landscape assessment results were compared according to visual reproduction methods. To investigate perceived affective soundscape and landscape qualities, the main perceptual components were extracted by conducting PCA on semantic expressions. Spearman rho correlations (r_s) were analyzed to examine correlations among all descriptor responses, including derived components, and correlations were compared according to visual reproduction methods.

3. Results and discussion

3.1. Subjective responses

3.1.1. Overall satisfaction and VR environment quality

Fig. 3 displays the assessments of mean overall satisfaction (O) and VR reproduction quality (R) for both visual environment reproduction methods. The mean rating scores of participant responses ranged from 1.85 to 4.38, depending on the site, indicating a diverse range of positive and negative assessments. Fig. 3 reveals that although there is a difference in the response mean, the response pattern is somewhat similar. As a result of examining the consistency between participant responses in two visual environments using rank correlation (Spearman's rho coefficient, r_s), weak but statistically significant relationship was derived for "overall impression" ($r_s = 0.19$, $p < 0.05$), "acoustic satisfaction" ($r_s = 0.18$, $p < 0.05$), "visual aesthetic" ($r_s = 0.19$, $p < 0.05$), "comfort" ($r_s = 0.35$, $p < 0.01$), and "immersion" ($r_s = 0.23$, $p < 0.01$). According to the Mann–Whitney *U* test results, the responses for both visual environment reproduction methods showed statistically significant differences for all descriptors. The mean subjective rating scores were higher in the monitor environment than the HMD by 0.33 (overall impression), 0.86 (acoustic satisfaction), 0.31 (visual aesthetic), 0.47 (comfort), 0.61 (realism), and 0.50 (immersion). Among the descriptors related to overall satisfaction (O), the mean difference in acoustic satisfaction was the highest, indicating that the visual environment affected this descriptor the most. The causes of these differences in the results of overall satisfaction according to the visual environment reproduction methods are briefly discussed here in consideration of the results presented in the later section. In terms of acoustic satisfaction, the assessment was more positive in the monitor environment due to the effect of traffic noise and machine noise, which are relatively perceived to be high in the HMD environment. In particular, this trend was most pronounced at Site D. In terms of visual aesthetic, the assessment was more positive in the monitor environment due to the effect of vehicle perceived to be relatively high in the HMD environment and the effect of vegetation perceived to be relatively high in the monitor environment. These differences in acoustic satisfaction and visual aesthetic were also reflected in overall impression and comfort, indicating that the assessment result was more negative in HMD environment in comparison. More sites were negatively assessed (given 3 on the Likert scale, representing neither bad nor good, or below) in the HMD environment than in the monitor. The mean VR reproduction quality (R) was at least 4 on the 7-point Likert scale, indicating that both visual methods had demonstrated somewhat high realism and immersion. However, the monitor reflected better reality judgment and received more positive satisfaction feedback, even in the same space.

3.1.2. Perceived audio-visual elements

Fig. 4 illustrates the audio-visual elements perceived according to visual environment reproduction methods. As different contexts depend on the space type and its function, a diverse range of audio-visual elements were also perceived at each site. Sound element perceptions showed that traffic and human sounds were perceived most dominantly overall (Fig. 4(a)). Traffic was generally perceived less strongly when further away from Site A, where a big road was located, while natural sounds, like birdsongs, were heard in some green spaces (i.e., Site F). Mann–Whitney *U* test results showed that all the sound elements had statistically significant differences depending on visual reproduction methods. Sound elements were perceived more dominantly in the HMD environment than in the monitor one, showing higher mean subjective rating scores in the order of 0.63, 0.89, 0.81, 0.60, 0.74, and 0.79, respectively, per site. Particularly, the difference between human sound perception in HMD and monitor environments was the highest (over 0.8) indicating that the visual environments affected this sound the most.

Visual element perceptions showed that various element combinations were perceived due to the different urban contexts (Fig. 4(b)). Along with sound elements, vehicles and roads were highly correlated with traffic, (0.77, $p < 0.01$) and (0.61, $p < 0.01$), respectively [77], showing similar response patterns depending on the site. However, people and human sounds, were not significantly correlated and neither were vegetation and birdsongs, indicating that perceptions of correlated visual and sound elements were not always consistent. The Mann–Whitney *U* test results showed that the visual elements presented statistically significant differences depending on visual reproduction methods. Among the visual elements, the vehicle category was 0.44 higher in the HMD environment than in the monitor, whereas sidewalk, high building, people, vegetation, and sky were, respectively, 0.50, 0.56, 0.55, 0.86, and 0.21 higher in the monitor environment. Thus, the visual elements showed different degrees of visual perception depending on the assessment environment. In summary, the visual elements were perceived more sensitively in the monitor, while the HMD environment received more sensitive responses to sound elements.

3.1.3. Perceived affective quality

To extract the main soundscape and landscape perceptual components, PCA was performed on the semantic expression test results (Table 2). We used 8 and 11 affective attributes for soundscape and landscape, respectively, and the mean subject responses were compared according to visual environment reproduction methods (Appendix B). A Varimax rotation was applied to determine the orthogonal components in PCA, wherein the Eigenvalue was set to >1. The Kaiser–Meyer–Olkin test results were 0.70 and 0.83 for soundscape and landscape, respectively, indicating adequate sampling. The Bartlett's sphericity test result was clear, confirming that the soundscape (Chi-squared: χ^2 (28) = 1,325.18, $p < 0.01$) and landscape (Chi-squared: χ^2 (55) = 1,540.66, $p < 0.01$) response data were suitable for PCA.

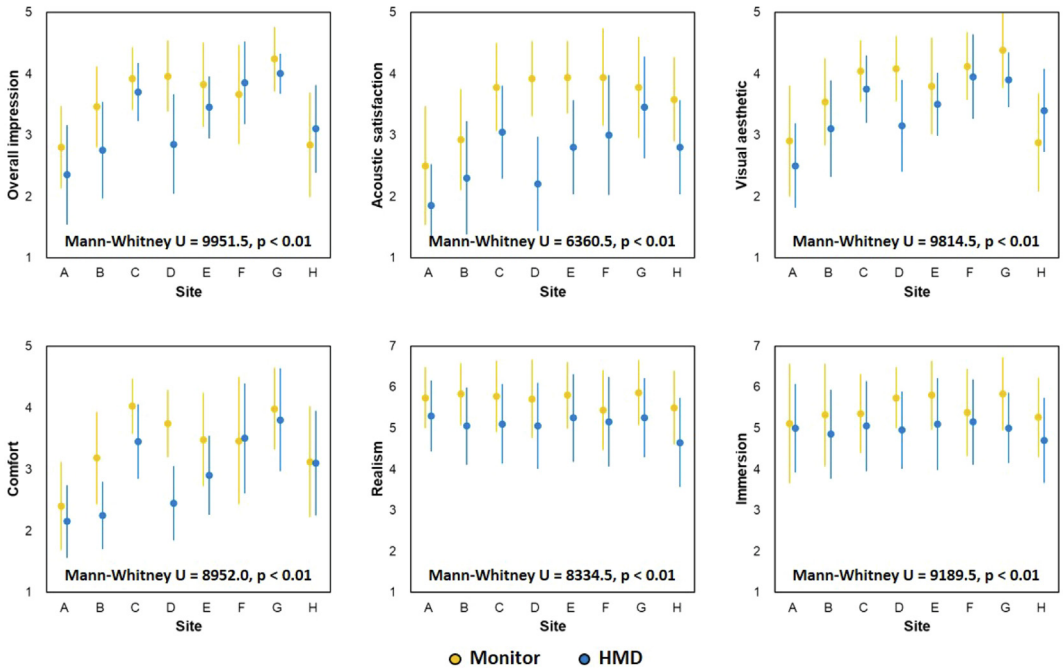

Fig. 3. Comparisons of six descriptors for monitor and HMD environments at eight sites. The corresponding *U* and *p* values of Mann–Whitney *U* tests are shown.

The soundscape perceptual components were classified as Component 1 or 2. Component 1 showed a 36.44% explanatory power and was highly correlated with positive attributes, such as eventful, exciting, and pleasant, representing a supportive soundscape. Component 2 had a 32.61% explanatory power, similar to Component 1, and was highly correlated with negative attributes, such as unpleasant and chaotic, corresponding to a detrimental soundscape. This is different from the commonly known pleasantness–eventfulness model in soundscape recognition [53]. These results, supportive and detrimental soundscapes, are consistent with Aletta et al.'s [76] categorizations of soundscape assessments through cluster analysis. The landscape perceptual components were classified into three components. Component 1 had a 30.71% explanatory power and high loading with negative attributes, such as repulsive, complex, and disharmonious. Therefore, it could be expressed as an annoying landscape. Component 2 showed a 21.02% explanatory power and was correlated with interesting, potentially representing an appealing landscape. Component 3 revealed a 14.10% explanatory power and was correlated with spatial attributes, such as enclosed and open, representing a surrounded landscape. These findings are similar to the results of previous research: Jeon & Hong [72] chose attractiveness, simplicity, and enclosure as the main perceptual factors of urban landscapes, and Jeon & Jo [17] suggested overall quality, regularity, spatial impression, and naturalness.

Factor scores are the values of multiple variables belonging to a single factor converted into a single score calculated through regression analysis [78]. We calculated the factor scores for the main PCA-derived soundscape and landscape perceptual components (Fig. 5). Mann–Whitney *U* test results showed that detrimental soundscape, annoying landscape, and surrounded landscape were significantly different depending on visual environment reproduction methods. These components showed higher mean factor scores in the HMD environment than in the monitor by 1.00, 0.60, and 0.56, respectively. Detrimental soundscape showed the highest mean difference among all the responses, with a maximum difference of 1.86 at Site D, indicating the visual environment reproduction methods affected this soundscape the most. In summary, within the different reproduction methods used to assess urban environments, the visual spatial perceptions and the negative perceptions of visual and auditory components all varied.

3.2. Comparisons between two visual reproduction methods

3.2.1. Different relationships among subjective descriptors

Thus far, we only examined the correlations among descriptors according to visual environment reproduction methods. In this section, we compare the correlations between five PCA-derived components (descriptors) and six descriptors assessed in monitor and HMD environments. Table 3 presents the results. The correlations among overall impression, acoustic satisfaction, visual aesthetic, comfort, detrimental soundscape, and annoying landscape were all significant in both visual environments, and all the correlation directions matched well ($p < 0.01$). However, the correlation sizes were slightly different among the six descriptors, depending on the visual environments. The relationship between comfort–acoustic satisfaction and comfort–visual aesthetic implied that the participants were affected more by the visual sensory stimuli in the monitor and the auditory sensory stimuli in the HMD environment when determining the comfort levels of the urban environment. Moreover, the overall impression–annoying landscape, acoustic satisfaction–detrimental soundscape, and visual aesthetic–detrimental soundscape relationships were clearly different in the monitor and HMD environments, suggesting that the participants were affected more by the characteristics of negative soundscape and landscape perceptions when determining overall satisfaction with urban environments, a tendency that appeared to be more frequent in the HMD environment. These findings are consistent with Jeon et al.'s [79] research findings, wherein the emotional responses to indoor noises were more prominent in the HMD environment than in laboratory settings, and the noises increased negative emotions.

Interestingly, among other descriptors, aside from the aforementioned six, positively perceived affective qualities, such as supportive soundscape and appealing landscape, were significantly correlated with other descriptors only in the monitor. In previous

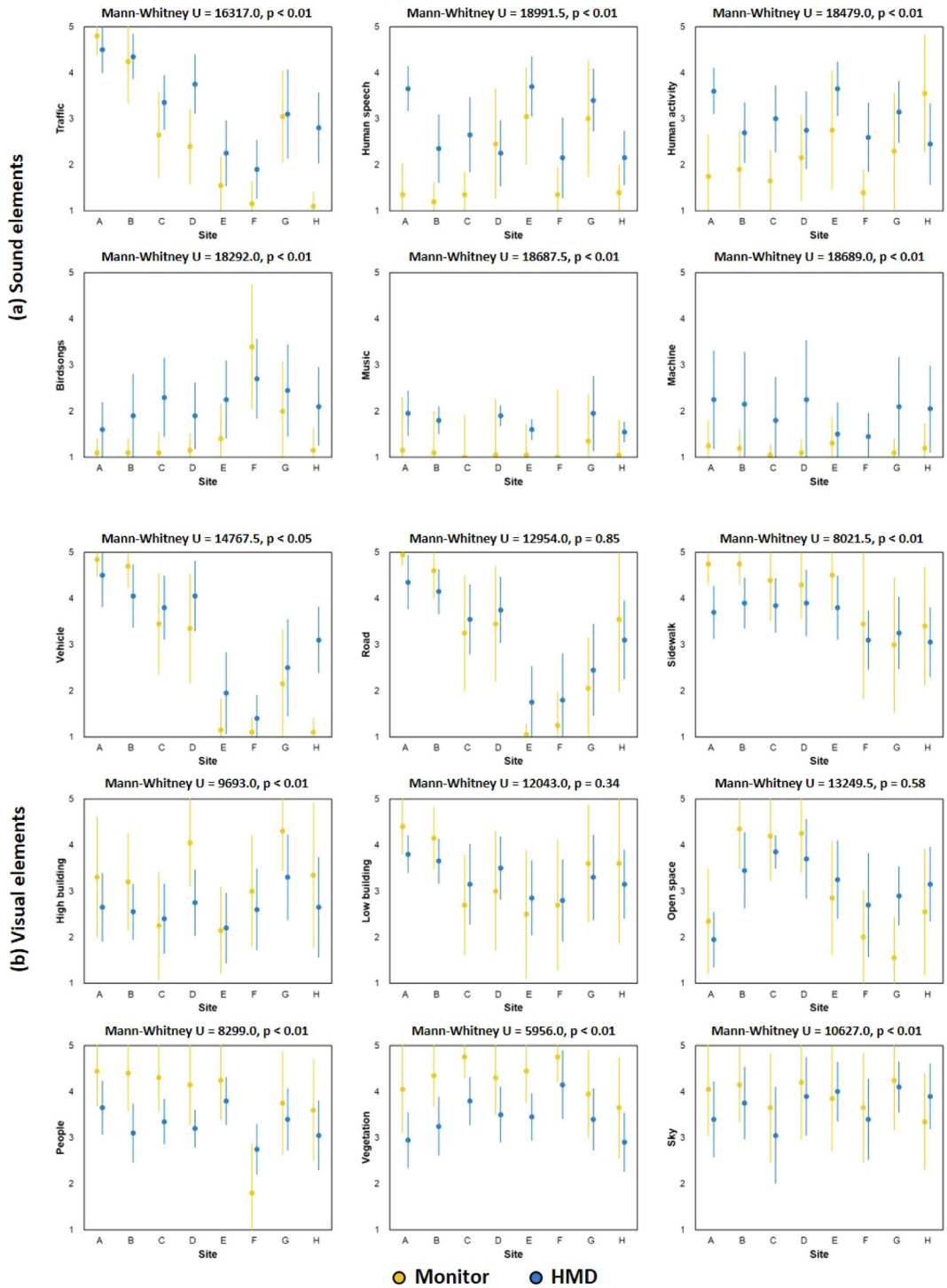

Fig. 4. Comparison of audio-visual elements consisting of (a) six sound and (b) nine visual elements for monitor and HMD environments at eight sites. The corresponding *U* and *p* values of Mann–Whitney *U* tests are shown.

studies [80,81], the HMD environment showed more undesirable effects (mental workload, physical discomfort, and simulation sickness) than the monitors, as well as higher emotional engagement and difficulties in feeling presence. Therefore, correlations with positive perceptual components may not have readily appeared due to these negative effects. Accordingly, the VR environment quality (R) related descriptors (immersion and realism) were also positively correlated with all overall satisfaction descriptors (O) only in the monitor environment. According to Azevedo et al. [82], this may be because realism and immersion decrease due to the visual distortion that occurs during the 360° image processing required to implement the HMD environment. Indeed, since the same image is also provided on the monitor, the distortion problem occurs in the same way, but the participants responded more sensitively to the distortion in the highly immersive HMD environment. This was expected as visual rendering algorithm technology must be further developed to achieve more accurate and realistically emotive VR environments.

Table 2
Rotated-PCA-component matrices obtained for semantic expressions. The numbers in parentheses represent explained variances.

Sound Attributes	Component 1 (36.44)	Component 2 (32.61)	Visual Attributes	Component 1 (30.71)	Component 2 (21.02)	Component 3 (14.10)
Supportive soundscape			***Annoying landscape***			
Eventful	0.87	0.07	Attractive	−0.55	0.58	−0.02
Uneventful	−0.82	0.02	Comfortable	−0.75	0.27	−0.12
Exciting	0.77	−0.06	Harmonious	−0.69	0.31	−0.10
Monotonous	−0.78	−0.04	Complex	0.76	0.22	0.04
			Repulsive	0.77	−0.17	0.23
Detrimental soundscape			Disharmonious	0.74	−0.36	0.17
Pleasant	0.51	−0.55	***Appealing landscape***			
Unpleasant	−0.05	0.88	Uninteresting	0.36	−0.72	0.12
Calm	−0.05	−0.86	Interesting	−0.32	0.78	−0.15
Chaotic	0.10	0.88	Simple	−0.18	−0.65	0.28
			Surrounded landscape			
			Open	−0.20	0.17	−0.79
			Enclosed	0.12	−0.15	0.84

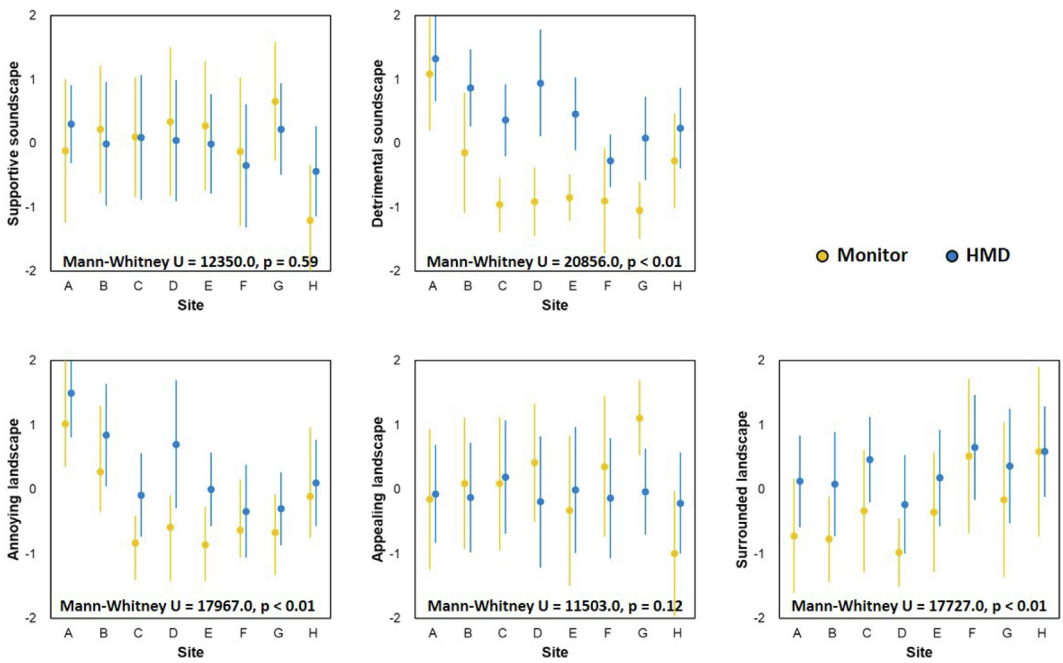

Fig. 5. Comparison of five components for monitor and HMD environments at eight sites. The corresponding *U* and *p* values of Mann–Whitney *U* tests are shown.

Furthermore, surrounded landscape showed contrasting correlations with overall satisfaction (O) descriptors depending on the reproduction methods: it was weakly negatively correlated with overall satisfaction (O) descriptors in the monitor environment and weakly positively correlated with them in the HMD. In other words, the enclosed spatial feeling was negative in the monitor environment and positive in the HMD. This perception gap was likely owing to the first/third person perspective (1/3PP) that varies depending on the visual reproduction method because environmental embodiment increases in 1PP (HMD), and this high-spatial-presence is a positive effect of HMD environments, among perhaps other positive [83–85]. However, using only the VR reproduction quality items presented in this study was insufficient to clearly distinguish between these effect possibilities. Thus, additional research is needed. Nonetheless, visual spatial characteristics, such as surrounded landscape, are important factors for assessing visual environment reproduction methods. Therefore, it is necessary to consider spatial characteristics related to changes in visual environment conditions as assessment factors for laboratory experiments.

3.2.2. Different effects of audio-visual elements on subjective assessments

The effects of perceived audio-visual elements on urban environmental perception were compared according to visual reproduction methods (Table 4). Artificial elements, such as traffic, vehicles, roads, high buildings, low buildings, and open spaces, negatively affected overall satisfaction (O) assessments and increased the negative soundscape and landscape perceptual components. Most elements (except vehicle and road) showed higher correlations in the HMD environment, thereby producing more negative effects. Although most elements somewhat positively affected positive components, most correlation sizes were <0.3. Therefore, the differences in correlations between elements in both visual environments were negligible. Meanwhile, natural elements (birdsongs, vegetation, and sky) not only positively affected overall satisfaction (O) but also reduced negative emotions and increased positive ones toward soundscape and landscape. These findings are consistent with results of previous studies [86,87], wherein natural elements positively affected soundscape and landscape evaluations. However, natural elements did not consistently show

Table 3
Comparisons of the descriptors in monitor and HMD environments. The lower left matrix shows the Spearman rho correlation coefficients obtained for different descriptors in the HMD environment. The upper right shows those obtained for different descriptors in the monitor environment (*$p < 0.05$, **$p < 0.01$).

Monitor	Overall impression	Acoustic satisfaction	Visual aesthetic	Comfort	Realism	Immersion	Supportive soundscape	Detrimental soundscape	Annoying landscape	Appealing landscape	Surrounded landscape
HMD											
Overall impression		0.48 **	0.78 **	0.61 **	0.32 **	0.35 **	0.37 **	−0.49 **	−0.44 **	0.46 **	−0.26 **
Acoustic satisfaction	0.64 **		0.55 **	0.45 **	0.17 *	0.31 **	0.21 **	−0.37 **	−0.42 **	0.27 **	0.02
Visual aesthetic	0.69 **	0.44 **		0.63 **	0.38 **	0.43 **	0.39 **	−0.55 **	−0.48 **	0.48 **	−0.22 **
Comfort	0.64 **	0.59 **	0.52 **		0.25 **	0.29 **	0.35 **	−0.60 **	−0.58 **	0.45 **	−0.09
Realism	0.08	0.05	0.07	0.21 **		0.71 **	0.25 **	−0.08	−0.07	0.24 **	−0.24 **
Immersion	0.10	0.01	0.11	0.23 **	0.80 **		0.26 **	−0.19 *	−0.13	0.27 **	−0.20 *
Supportive soundscape	−0.05	0.02	−0.07	0.11	0.11	0.10		−0.17 *	−0.08	0.75 **	−0.28 **
Detrimental soundscape	−0.54 **	−0.58 **	−0.40 **	−0.52 **	−0.01	−0.01	0.24 **		0.73 **	−0.31 **	0.06
Annoying landscape	−0.65 **	−0.36 **	−0.53 **	−0.59 **	−0.24 **	−0.23 **	0.13	0.53 **		−0.06	0.03
Appealing landscape	0.11	0.20 *	0.09	0.10	−0.07	−0.06	0.46 **	0.00	0.14		−0.12
Surrounded landscape	0.18 *	0.28 **	0.12	0.21 **	−0.12	−0.01	0.02	−0.31 **	−0.17 *	0.36 **	

Table 4
Spearman rho correlation coefficients obtained for correlations between audio-visual elements and subjective ratings in monitor and HMD visual environments (*$p < 0.05$, **$p < 0.01$).

Monitor	Overall impression	Acoustic satisfaction	Visual aesthetic	Comfort	Realism	Immersion	Supportive soundscape	Detrimental soundscape	Annoying landscape	Appealing landscape	Surrounded landscape
HMD											
Overall impression		0.48 **	0.78 **	0.61 **	0.32 **	0.35 **	0.37 **	−0.49 **	−0.44 **	0.46 **	−0.26 **
Acoustic satisfaction	0.64 **		0.55 **	0.45 **	0.17 *	0.31 **	0.21 **	−0.37 **	−0.42 **	0.27 **	0.02
Visual aesthetic	0.69 **	0.44 **		0.63 **	0.38 **	0.43 **	0.39 **	−0.55 **	−0.48 **	0.48 **	−0.22 **
Comfort	0.64 **	0.59 **	0.52 **		0.25 **	0.29 **	0.35 **	−0.60 **	−0.58 **	0.45 **	−0.09
Realism	0.08	0.05	0.07	0.21 **		0.71 **	0.25 **	−0.08	−0.07	0.24 **	−0.24 **
Immersion	0.10	0.01	0.11	0.23 **	0.80 **		0.26 **	−0.19 *	−0.13	0.27 **	−0.20 *
Supportive soundscape	−0.05	0.02	−0.07	0.11	0.11	0.10		−0.17 *	−0.08	0.75 **	−0.28 **
Detrimental soundscape	−0.54 **	−0.58 **	−0.40 **	−0.52 **	−0.01	−0.01	0.24 **		0.73 **	−0.31 **	0.06
Annoying landscape	−0.65 **	−0.36 **	−0.53 **	−0.59 **	−0.24 **	−0.23 **	0.13	0.53 **		−0.06	0.03
Appealing landscape	0.11	0.20 *	0.09	0.10	−0.07	−0.06	0.46 **	0.00	0.14		−0.12
Surrounded landscape	0.18 *	0.28 **	0.12	0.21 **	−0.12	−0.01	0.02	−0.31 **	−0.17 *	0.36 **	

different correlations with descriptors depending on the visual reproduction methods. Thus, the effects of perceived audio-visual elements could be divided into artificial and natural elements. Although these elements both positively affected urban contextual perceptions, the negative effects of artificial elements appeared to be more sensitively perceived in the HMD environment.

Interestingly, the surrounded landscape was negatively correlated with more audio-visual elements in the monitor, and elements, such as birdsongs and vehicles, showed conflicting complex correlations in different environments because, even in the same space, individual perceptions of audio-visual elements forming complicated urban contexts were affected by spatial presence, which varied depending on view size [88]. Moreover, according to Gorisse et al. [89], 3PP (monitor) is more suitable for perceiving projectile motions than 1PP (HMD), considering spatial awareness and environmental perception capacities. Accordingly, the results of this study showed that the participants determined spatial surrounded landscapes by comprehensively considering more types of audio-visual elements in the monitor (3PP). Therefore, using the monitor to assess urban contexts in laboratory settings enables responses to be obtained by comprehensively considering multiple components of the surrounding environment. However, it is also necessary to consider personal perspectives that are different from the actual environment.

3.2.3. Effects of acoustical parameters on subjective assessments

To compare the physical acoustical characteristics affecting subjective assessments according to visual environmental conditions, the correlations between acoustical parameters and descriptors were analyzed (Table 5). Spectral contents ($L_{Ceq\text{-}Aeq}$), sound strengths (L_{Aeq}), and psychoacoustic property (Loudness, Sharpness, Fluctuation strength, Roughness, Tonality) were significantly correlated with overall satisfaction (O), soundscape (S), and landscape (L) perceptual components ($p < 0.01$). $L_{Ceq\text{-}Aeq}$ correlated more with overall satisfaction (O) than L_{Aeq} and Loudness, suggesting that along with average SPL, the negative low-frequency component was a highly important characteristic in urban contextual perception. This relationship was also confirmed in Sharpness. Therefore, it was found that the influence on high frequency as well as low frequency characteristics of noise sources generated in the urban environment should be considered as important. Among the descriptors, although only acoustic satisfaction and surrounded landscape had more sensitive responses to acoustical characteristics in the monitor, most descriptors were more highly correlated with the acoustical parameters in the HMD, because the latter produced more sensitive responses to sound elements (Fig. 4).

Meanwhile, for the sound temporal variability ($L_{A10\text{-}A90}$), only acoustic satisfaction was significantly negatively correlated with overall satisfaction (O) in the monitor environment. This result is consistent with Jo and Jeon's [90] study, in which the participants became desensitized to changes in sound characteristics, like discontinuity, in the HMD environment. However, unlike $L_{A10\text{-}A90}$, which is an overall negative fluctuation index, the result of fluctuation strength, which means the fluctuation characteristics of low-frequency noise, showed a stronger correlation with overall satisfaction indicators within the HMD environment. In addition, a similar trend was confirmed in Roughness, and through this, it was reconfirmed that the subjects responded more sensitively to urban noise in the HMD environment. The soundscape and landscape perceptual components were weakly, yet more significantly, correlated than the spectral contents and overall sound strength in both visual environments. The surrounded landscape was negatively correlated with all acoustical characteristics only in the monitor, while in the HMD, it was significantly correlated with only $L_{Ceq\text{-}Aeq}$ and fluctuation strength. As previously discussed, the participants comprehensively considered more diverse audio-visual elements to assess spatial characteristics in the monitor, which explains why the sound elements related to physical sound characteristics were significantly correlated. Realism and immersion were not significantly correlated ($p > 0.05$), confirming that their acoustical effect on the VR environment (R) quality was negligible [28]. By placing urban sound environments in order of importance (low-frequency content > sound-pressure level > temporal variability) not only could satisfaction be increased, but also negative emotions toward soundscapes and landscapes could be reduced. In terms of psychoacoustic, it was confirmed that the urban sound environment needs to be improved with importance in the order of Fluctuation > Loudness and Sharpness > Roughness and Tonality. Although these urban design effects may appear to have a greater overall impact on the HMD, they were equally positive in both reproduction methods.

3.3. Overall comparisons of different analyses

Table 6 summarizes the results obtained from the subjective evaluations according to changes in monitor and HMD environments. The mean responses for overall impression, acoustic satisfaction, visual aesthetic, comfort, realism, immersion, detrimental soundscape, annoying landscape, and surrounded landscape were different in both environments. The descriptors related to overall satisfaction (O) and VR reproduction quality (R) were highly correlated in the monitor, whereas the rest were highly correlated in the HMD. Interestingly, although the mean comfort may be different in each visual environment, it showed the most similar tendency in both reproduction methods when correlated with audio-visual elements, descriptor comparisons, and acoustical parameters (Table 6). Therefore, comfort is a suitable assessment descriptor. Except for the latter descriptor, although the monitor showed more correlations between other descriptors than the HMD, indicators, such as audio-visual elements and acoustical parameters, affected descriptors differently depending on the visual reproduction method, making it difficult to determine whether certain visual conditions were always correct.

Spatial attributes showed opposite correlations in the HMD and monitor environments. They were correlated with more visual elements and acoustical parameters in the monitor, possibly because realism and immersion were highly correlated with the VR environment spatial presence. Therefore, it is necessary to consider visual spatial aspects when assessing urban environment descriptors. Although positive perceptual components, such as supportive soundscape and appealing landscape, showed similar mean responses depending on the visual environment reproduction methods, consistency among the responses was low in both visual environments. Furthermore, other descriptors and audio-visual elements were better correlated only in the monitor. Acoustical parameters, however, were better correlated only in the HMD. Therefore, perceptions of urban contexts are clearly different depending on how visual data are provided, suggesting that researchers must sufficiently consider all influencing factors when analyzing urban design quality in laboratory settings.

3.4. Contributions, limitations, and future study

The distinctive perception results presented new points to consider for selecting visual environment reproduction methods to develop VR environments in laboratory settings. Our discoveries and data could be extensively used as a basis for developing experimental methodologies to assess the soundscape and landscape of urban environments in laboratory settings. This study introduced emotive VR assessment environments, providing crucial considerations for planning and analyzing audio-visual perceptions of

Table 5
Spearman's rho correlation coefficients between acoustic and psycho-acoustic parameters and subjective ratings of two visual environments. (*p < 0.05, **p < 0.01).

Descriptor		L_{Aeq}		$L_{A10-A90}$		$L_{Ceq-Aeq}$		Loudness		Sharpness		FS		Ro		To	
Overall impression	Monitor	−0.16	*	−0.05		−0.21	**	−0.16	*	−0.31	**	−0.11		0.02		−0.35	**
	HMD	−0.34	**	−0.06		−0.37	**	−0.26	**	−0.35	**	−0.38	**	−0.21	**	−0.18	*
Acoustic satisfaction	Monitor	−0.41	**	−0.30	**	−0.39	**	−0.24	**	−0.29	**	−0.33	**	−0.20	*	−0.22	**
	HMD	−0.25	**	−0.10		−0.26	**	−0.33	**	−0.36	**	−0.37	**	−0.22	**	−0.21	**
Visual aesthetic	Monitor	−0.20	*	−0.03		−0.24	**	−0.06		−0.21	**	−0.08		0.05		−0.26	**
	HMD	−0.34	**	−0.14		−0.37	**	−0.27	**	−0.31	**	−0.36	**	−0.25	**	−0.17	*
Comfort	Monitor	−0.18	*	−0.11		−0.33	**	−0.26	**	−0.33	**	−0.20	*	−0.06		−0.37	**
	HMD	−0.33	**	−0.11		−0.40	**	−0.33	**	−0.36	**	−0.43	**	−0.21	**	−0.21	**
Realism	Monitor	0.10		0.08		0.09		0.00		−0.04		0.06		0.06		−0.06	
	HMD	0.03		0.12		0.06		0.09		0.02		0.07		0.06		0.03	
Immersion	Monitor	−0.06		−0.07		0.04		−0.06		−0.12		−0.05		0.01		−0.11	
	HMD	−0.03		0.05		−0.05		0.06		0.02		0.02		0.00		0.03	
Supportive soundscape	Monitor	0.09		0.17	*	0.11		0.10		−0.06		0.16	*	0.17	*	−0.12	
	HMD	0.20	**	0.20	*	0.16	*	0.09		0.02		0.18	*	0.20	*	−0.07	
Detrimental soundscape	Monitor	0.37	**	0.21	**	0.43	**	0.23	**	0.32	**	0.28	**	0.15		0.30	**
	HMD	0.45	**	0.22	**	0.40	**	0.24	**	0.24	**	0.43	**	0.33	**	0.03	
Annoying landscape	Monitor	0.38	**	0.24	**	0.40	**	0.31	**	0.38	**	0.36	**	0.26	**	0.28	**
	HMD	0.45	**	0.24	**	0.44	**	0.36	**	0.39	**	0.49	**	0.35	**	0.17	*
Appealing landscape	Monitor	0.05		0.15		0.03		0.07		−0.07		0.11		0.24	**	−0.19	*
	HMD	0.02		0.08		−0.03		−0.01		−0.04		0.00		−0.01		−0.02	
Surrounded landscape	Monitor	−0.29	**	−0.16	*	−0.24	**	−0.17	*	−0.09		−0.34	**	−0.29	**	0.12	
	HMD	−0.12		0.00		−0.17	*	−0.09		−0.04		−0.18	*	−0.15		0.07	

Note. FS: Fluctuation strength, Ro: Roughness, To: Tonality.

Table 6
Summary of different analyses of monitor and HMD environments (*p < 0.05, **p < 0.01).

Groups	Descriptors	Rating differences [a]		Sound elements (h/6) [b]		Visual elements (i/9) [c]		Descriptor comparisons (j/10) [d]		Acoustical parameters (k/8) [e]	
		U	p	Monitor	HMD	Monitor	HMD	Monitor	HMD	Monitor	HMD
O	*Overall impression*	9951.5	< 0.01	1/6*	1/6**	3/9**	4/9**, 1/9*	10/10**	5/10**, 1/10*	3/8**, 2/8*	6/8**, 1/8
	Acoustic satisfaction	6360.5	< 0.01	2/6**, 2/6*	1/6**, 1/6*	3/9**, 1/9*	3/9**, 1/9*	8/10**, 1/10*	6/10**, 1/10*	7/8**, 1/8*	7/8**
	Visual aesthetic	9814.5	< 0.01	2/6**, 1/6*	1/6**	4/9**	4/9**, 1/9*	10/10**	5/10**	3/8**, 1/8*	6/8**, 1/8*
	Comfort	8952.0	< 0.01	1/6*	1/6**	4/9**	3/9**, 1/9*	9/10**	8/10**	4/8**, 2/8*	7/8**
R	*Realism*	8334.5	< 0.01	0/6	1/6**	6/9**, 2/9*	1/9**, 3/9*	7/10**, 1/10*	3/10**	0/8	0/8
	Immersion	9189.5	< 0.01	1/6*	0/6	4/9**, 2/9*	2/9*	7/10**, 2/10*	3/10**	0/8	0/8
S	*Supportive soundscape*	12350.0	0.56	2/6**, 2/6*	1/6**, 1/6*	4/9*	1/9**, 2/9*	8/10**, 1/10*	2/10**	3/8*	1/8**, 4/8*
	Detrimental soundscape	20856.0	< 0.01	2/6**, 2/6*	2/6**	4/9**, 1/9*	4/9**, 1/9*	6/10**, 2/10*	7/10**	7/8**	7/8**
L	*Annoying landscape*	17967.0	< 0.01	3/6**	2/6**	4/9**	4/9**	5/10**	7/10**, 1/10*	8/8**	7/8**, 1/8*
	Appealing landscape	11503.0	0.12	1/6**	1/6*	1/9**, 1/9*	0/9	8/10**	2/10**, 1/10*	1/8**, 1/8*	0/8
	Surrounded landscape	17727.0	< 0.01	1/6**, 1/6*	1/6**, 1/6*	6/9**, 1/9*	2/9**, 2/9*	4/10**, 1/10*	4/10**, 2/10*	4/8**, 2/8*	2/8*

[a] U and p are obtained from Mann–Whitney U tests for both reproduction methods.
[b] n is the number of r_s showing statistically significant correlations between both methods at eight sites.
[c] h is the number of r_s showing statistically significant correlation between subjective ratings and six sound elements.
[d] i is the number of r_s showing statistically significant correlation between subjective ratings and nine sound elements.
[e] j is the number of r_s showing statistically significant correlations between the other 10 descriptors and their comparisons.
[f] k is the number of r_s showing statistically significant correlations between subjective ratings and eight acoustical parameters.

urban contexts and for subjective assessment studies. However, it has the following limitations. First, visual environments were divided into separate HMD and monitor displays that were subsequently compared. However, by performing the same assessment in situ and comparing the current results to those obtained in actual environments, the utility of the study could be increased to design and plan urban environments. Moreover, this study does not represent other products showing similar functions in hardware and software performances. Therefore, it is necessary to conduct comparative experiments on various HMD and monitor models in subsequent studies. In addition, it is also necessary to compare various methodologies, such as the Cave system, using a large screen, etc., in addition to these two visual projection methods. Furthermore, the results cannot be generalized to all ages, because most of the participants were in their twenties. In addition, although this study performed assessment with participants of a small group with unbalanced male/female ratio, it is necessary to derive generalized interpretations of the results through conducting research with a larger group with a balanced gender ratio in future studies [91]. As this study assessed the VR reproduction quality, implemented using HMD and monitor environments, merely through immersion and realism, further research must investigate various psychological/physical quantitative indicators to measure the VR environment quality [84,92,93]. Such research would enable an in-depth analysis of the effects of the visual reproduction methods on technological assessments of soundscape and landscape, in addition to the psychological assessments examined in this study.

4. Conclusions

This study examined the effects of visual environment reproduction methods on perceptions of soundscapes and landscapes in urban environments. It introduced emotive VR assessment envi-

ronments, providing crucial considerations for planning and analyzing audio-visual perceptions of urban contexts and for subjective assessment studies. Urban environments were more sensitively assessed through the HMD than in the monitor because, although the positive perceptual soundscape and landscape components were similar, the negative perceptual components were more clearly perceived in the HMD environment. Furthermore, the artificial audio-visual elements affected urban environment perceptions more negatively in the HMD environment, and the effects of natural elements were almost negligible in both visual environments. The monitor had more significant correlations among descriptors and more-diverse correlations among audio-visual elements, suggesting that it offers higher complex awareness of urban contexts. Perceptions of landscape spatial attributes were positively and negatively affected in HMD and monitor environments, respectively. As visual environments affected this attribute the most, it is necessary to consider the effects of spatial aspects on visual reproduction methods. The most important physical characteristics of sounds that affect urban contextual perceptions were spectral contents, followed by sound strength and temporal variation, all of which showed similar tendencies in both visual environments. Overall, perception of audio-visual elements, responses to negative or positive emotions, and general awareness of urban contexts all varied depending on visual environments. Monitor and HMD environments showed unique strengths and weaknesses, benefiting awareness and spatial presence, respectively. Therefore, a suitable visual environment must be selected depending on the research purpose. The data and discoveries of this study can be widely used as a basis for developing methodologies that assess urban environment soundscapes and landscapes in laboratory settings. Moreover, they can help urban design and planning by promoting understandings of urban contexts and audio-visual interactions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A.

Table A1.

Appendix B.

Fig. A1.

Table A1
Questionnaire details.

	Questions	5-Point Likert Scale				
		1	2	3	4	5
O	To what extent is the OOO of the present surrounding environment? - overall impression, acoustic satisfaction, visual aesthetic, comfort	very bad	bad	neither bad nor good	good	very good
I	To what extent do you currently hear the following types of sound? - traffic, human speech, human activity, birdsongs, music, machines	not at all	a little	moderately	a lot	sound dominates
	To what extent do you currently see the following types of visual elements? - vehicles, roads, sidewalks, high buildings, low buildings, open spaces, people, vegetation, sky	not at all	a little	moderately	a lot	vision dominates
S	To what extent do the eight attributes below describe the present sound environment? - pleasant, unpleasant, eventful, uneventful, exciting, monotonous, calm, chaotic	strongly disagree	slightly disagree	neither disagree nor agree	slightly agree	strongly disagree
L	To what extent do the 12 attributes below describe your experience of the present visual environment? - attractive, comfortable, uninteresting, harmonious, complex, open, repulsive, interesting, enclosed, simple, disharmonious	strongly disagree	slightly disagree	neither disagree nor agree	slightly agree	strongly disagree

		7-Point Likert Scale						
		1	2	3	4	5	6	7
R	To what extent is the OOO of the present surrounding environment? - realism, immersion	not at all	very little	a little	moderately	fairly	very	extremely

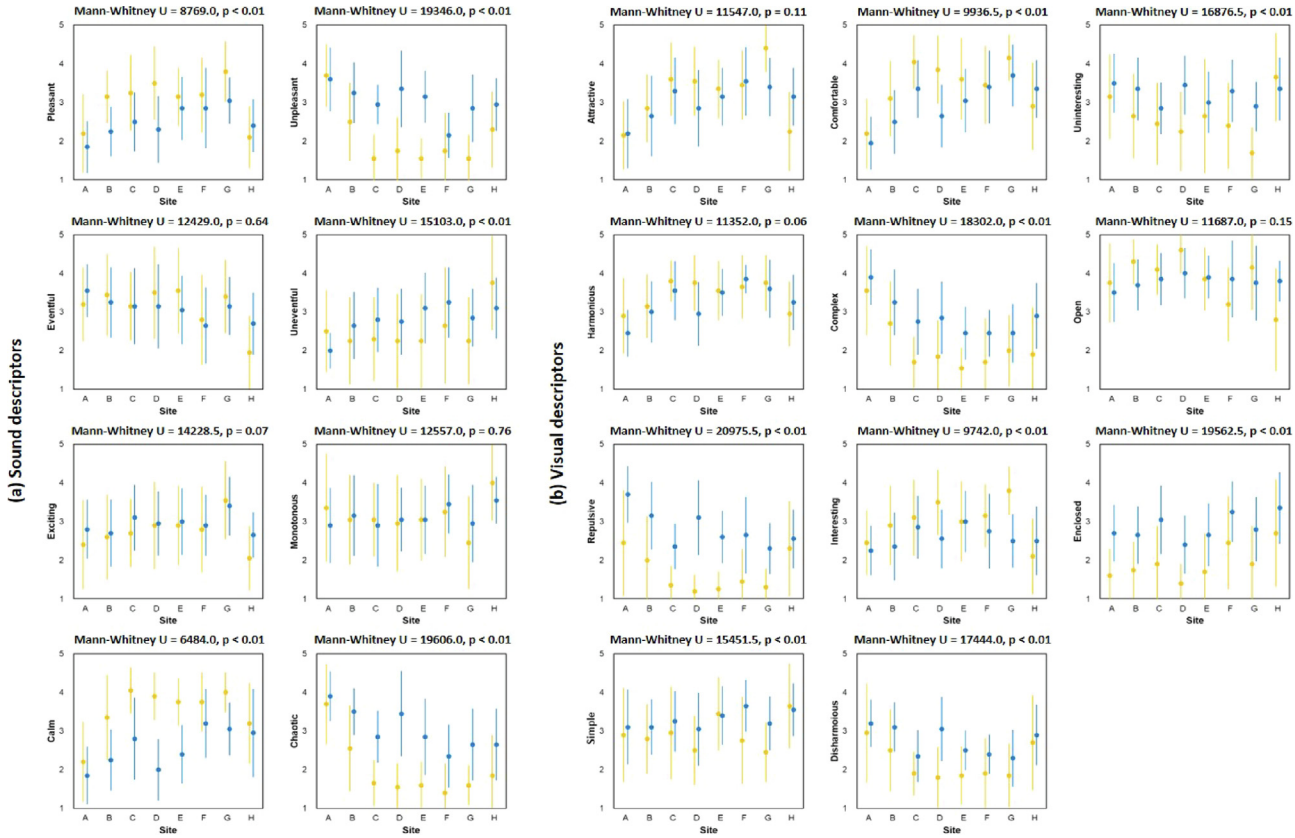

Fig. A1. Comparison of eight sound-descriptor attributes and 11 visual-descriptor attributes obtained for monitor and HMD environments at eight sites. Corresponding *U* and *p* values obtained from Mann–Whitney *U* tests are listed.

References

[1] Orland B, Budthimedhee K, Uusitalo J. Considering virtual worlds as representations of landscape realities and as tools for landscape planning. Landscape Urban Plann 2001;54(1–4):139–48. https://doi.org/10.1016/S0169-2046(01)00132-3.

[2] Parsons TD. Virtual reality for enhanced ecological validity and experimental control in the clinical, affective and social neurosciences. Front Hum Neurosci 2015;9:660. https://doi.org/10.3389/fnhum.2015.00660.

[3] Pitt DG, Nassauer JI. Virtual reality systems and research on the perception, simulation and presentation of environmental change. Landscape Urban Plann 1992;21(4):269–71. https://doi.org/10.1016/0169-2046(92)90038-2.

[4] Schmuckler MA. What is ecological validity? A dimensional analysis. Infancy 2001;2(4):419–36. https://doi.org/10.1207/S15327078IN0204_02.

[5] Burdea GC, Coiffet P (Eds.). (2003). *Virtual reality technology*. Hoboken: John Wiley & Sons. https://doi.org/10.1162/105474603322955950.

[6] Heydarian A, Carneiro JP, Gerber D, Becerik-Gerber B, Hayes T, Wood W. Immersive virtual environments versus physical built environments: A benchmarking study for building design and user-built environment explorations. Autom Constr 2015;54:116–26. https://doi.org/10.1016/j.autcon.2015.03.020.

[7] Zacharias J. Exploratory spatial behaviour in real and virtual environments. Landscape Urban Plann 2006;78(1–2):1–13. https://doi.org/10.1016/j.landurbplan.2005.05.002.

[8] Freina L, Ott M. A literature review on immersive virtual reality in education: state of the art and perspectives. *The International Scientific Conference eLearning and Software for Education*. Bucharest, Romania, Carol I National Defense University. Retrieved from: https://www.researchgate.net/publication/280566372_A_Literature_Review_on_Immersive_Virtual_Reality_in_Education_State_Of_The_Art_and_Perspectives/link/55ba24b208ae9289a0926382/download. (2015).

[9] Weiss PL, Kizony R, Feintuch U, Katz N, Selzer M, Clarke S, et al. Virtual reality in neurorehabilitation. In: Selzer M, Clarke S, Cohen L, Duncan P, Gage F, editors. Textbook of neural repair and rehabilitation. Cambridge: Cambridge University Press; 2006. p. 182–97. https://doi.org/10.1017/CBO9780511545078.015.

[10] Axford S, Keltie G, Wallis C. Virtual reality in urban planning and design. In: Cartwright W, Peterson MP, Gartner G, editors. Multimedia cartography. Berlin, Heidelberg: Springer; 2007. p. 283–94. https://doi.org/10.1007/978-3-540-36651-5_20.

[11] Birenboim A, Dijst M, Ettema D, de Kruijf J, de Leeuw G, Dogterom N. The utilization of immersive virtual environments for the investigation of environmental preferences. Landscape Urban Plann 2019;189:129–38. https://doi.org/10.1016/j.landurbplan.2019.04.011.

[12] Bruns CR, Chamberlain BC. The influence of landmarks and urban form on cognitive maps using virtual reality. Landscape Urban Plann 2019;189:296–306. https://doi.org/10.1016/j.landurbplan.2019.05.006.

[13] Meffei L, Masullo M, Pascale A, Ruggiero G, Romero VP. On the validity of immersive virtual reality as tool for multisensory evaluation of urban spaces. Energy Procedia 2015;78:471–6. https://doi.org/10.1016/j.egypro.2015.11.703.

[14] Natapov A, Fisher-Gewirtzman D. Visibility of urban activities and pedestrian routes: An experiment in a virtual environment. Comput Environ Urban Syst 2016;58:60–70. https://doi.org/10.1016/j.compenvurbsys.2016.03.007.

[15] Carles J, Bernáldez F, Lucio JD. Audio visual interactions and soundscape preferences. Landscape Res 1992;17(2):52–6. https://doi.org/10.1080/01426399208706361.

[16] Hong JY, Jeon JY. The effects of audio-visual factors on perceptions of environmental noise barrier performance. Landscape Urban Plann 2014;125:28–37. https://doi.org/10.1016/j.landurbplan.2014.02.001.

[17] Jeon JY, Jo HI. Effects of audio-visual interactions on soundscape and landscape perception and their influence on satisfaction with the urban environment. Build Environ 2020;169:106544. https://doi.org/10.1016/j.buildenv.2019.106544.

[18] Yu T, Behm H, Bill R, Kang J. Audio-visual perception of new wind parks. Landscape Urban Plann 2017;165:1–10. https://doi.org/10.1016/j.landurbplan.2017.04.012.

[19] Jo HI, Jeon JY. Effect of the appropriateness of sound environment on urban soundscape assessment. Build Environ 2020;179:106975. https://doi.org/10.1016/j.buildenv.2020.106975.

[20] Drettakis G, Roussou M, Reche A, Tsingos N. Design and evaluation of a real-world virtual environment for architecture and urban planning. Presence: Teleoper Virtual Environ 2007;16(3):318–32. https://doi.org/10.1162/pres.16.3.318.

[21] Ruotolo F, Maffei L, Di Gabriele M, Iachini T, Masullo M, Ruggiero G, et al. Immersive virtual reality and environmental noise assessment: An innovative audio-visual approach. Environ Impact Assess Rev 2013;41:10–20. https://doi.org/10.1016/j.eiar.2013.01.007.

[22] Sanchez GME, Van Renterghem T, Sun K, De Coensel B, Botteldooren D. Using Virtual Reality for assessing the role of noise in the audio-visual design of an

urban public space. Landscape Urban Plann 2017;167:98–107. https://doi.org/10.1016/j.landurbplan.2017.05.018.
[23] Maffei L, Masullo M, Pascale A, Ruggiero G, Romero VP. Immersive virtual reality in community planning: Acoustic and visual congruence of simulated vs real world. Sustain Cities Soc 2016;27:338–45. https://doi.org/10.1016/j.scs.2016.06.022.
[24] ISO/DIS 12913-2. Acoustics. Soundscape-part 2: Data collection and reporting requirements. Retrieved from: https://www.iso.org/standard/75267.html. (2018).
[25] Hong JY, Lam B, Ong Z-T, Ooi K, Gan W-S, Kang J, et al. Quality assessment of acoustic environment reproduction methods for cinematic virtual reality in soundscape applications. Build Environ 2019;149:1–14. https://doi.org/10.1016/j.buildenv.2018.12.004.
[26] Guastavino C, Katz BF. Perceptual evaluation of multi-dimensional spatial audio reproduction. J Acoust Soc Am 2004;116(2):1105–15. https://doi.org/10.1121/1.1763973.
[27] Guastavino C, Katz BF, Polack JD, Levitin DJ, Dubois D. Ecological validity of soundscape reproduction. Acta Acustica United with Acustica 2005;91(2):333–41.
[28] Xu C, Kang J. Soundscape evaluation: Binaural or monaural? J Acoust Soc Am 2019;145(5):3208–17. https://doi.org/10.1121/1.5102164.
[29] Hong JY, Lam B, Ong Z-T, Ooi K, Gan W-S, Kang J, et al. The effects of spatial separations between water sound and traffic noise sources on soundscape assessment. Build Environ 2020;167:106423. https://doi.org/10.1016/j.buildenv.2019.106423.
[30] Juan MC, Pérez D. Comparison of the levels of presence and anxiety in an acrophobic environment viewed via HMD or CAVE. Presence: Teleoper Virtual Environ 2009;18(3):232–48. https://doi.org/10.1162/pres.18.3.232.
[31] Kim K, Rosenthal MZ, Zielinski D, Brady R. Comparison of desktop, head mounted display, and six wall fully immersive systems using a stressful task. *2012 IEEE virtual reality workshops* (*VRW*) (2012) (pp. 143-144). https://doi.org/10.1109/VR.2012.6180922.
[32] Lin CJ, Abreham BT, Woldegiorgis BH. Effects of displays on a direct reaching task: A comparative study of head mounted display and stereoscopic widescreen display. Int J Ind Ergon 2019;72:372–9. https://doi.org/10.1016/j.ergon.2019.06.013.
[33] Meyerbröker K, Morina N, Kerkhof G, Emmelkamp PM. Virtual reality exposure treatment of agoraphobia: A comparison of computer automatic virtual environment and head-mounted display. Annu Rev Cyberther Telemed 2011;9(1):41–5. https://doi.org/10.1016/j.scs.2016.06.022.
[34] Feng D, O'Halloran KL. Representing emotive meaning in visual images: A social semiotic approach. J Pragmat 2012;44(14):2067–84. https://doi.org/10.1016/j.pragma.2012.10.003.
[35] Lange E. The limits of realism: perceptions of virtual landscapes. Landscape Urban Plann 2001;54(1–4):163–82. https://doi.org/10.1016/S0169-2046(01)00134-7.
[36] Herbert G, Chen X. A comparison of usefulness of 2D and 3D representations of urban planning. Cartogr Geogr Inf Sci 2015;42(1):22–32. https://doi.org/10.1080/15230406.2014.987694.
[37] Jiang L, Masullo M, Maffei L, Meng F, Vorländer M. A demonstrator tool of web-based virtual reality for participatory evaluation of urban sound environment. Landscape Urban Plann 2018;170:276–82. https://doi.org/10.1016/j.landurbplan.2017.09.007.
[38] Lim EM, Honjo T, Umeki K. The validity of VRML images as a stimulus for landscape assessment. Landscape Urban Plann 2006;77(1–2):80–93. https://doi.org/10.1016/j.landurbplan.2005.01.007.
[39] Lim EM, Honjo T. Three-dimensional visualization forest of landscapes by VRML. Landscape Urban Plann 2003;63(3):175–86. https://doi.org/10.1016/S0169-2046(02)00189-5.
[40] Puyana-Romero V, Lopez-Segura LS, Maffei L, Hernández-Molina R, Masullo M. Interactive soundscapes: 360°-video based immersive virtual reality in a tool for the participatory acoustic environment evaluation of urban areas. Acta Acustica United with Acustica 2017;103(4):574–88. https://doi.org/10.3813/AAA.919086.
[41] Davies WJ, Bruce NS, Murphy JE. Soundscape reproduction and synthesis. Acta Acustica United with Acustica 2014;100(2):285–92. https://doi.org/10.3813/AAA.918708.
[42] Aletta F, Kang J, Axelsson Ö. Soundscape descriptors and a conceptual framework for developing predictive soundscape models. Landscape Urban Plann 2016;149:65–74. https://doi.org/10.1016/j.landurbplan.2016.02.001.
[43] Jeon JY, Jo HI. Three-dimensional virtual reality-based subjective evaluation of road traffic noise heard in urban high-rise residential buildings. Build Environ 2019;148:468–77. https://doi.org/10.1016/j.buildenv.2018.11.004.
[44] Slater M, Khanna P, Mortensen J, Yu I. Visual realism enhances realistic response in an immersive virtual environment. IEEE Comput Graphics Appl 2009;29(3):76–84. https://doi.org/10.1109/MCG.2009.55.
[45] Aburawis AAM, Dokmeci-Yorukoglu PN. An integrated framework on soundscape perception and spatial experience by adapting post-occupancy evaluation methodology. Build Acoust 2018;25(1):3–16. https://doi.org/10.1177/1351010X18758478.
[46] Zhou Z, Kang J, Jin H. Factors that influence soundscapes in historical areas. Noise Control Eng J 2014;62(2):60–8. https://doi.org/10.3397/1/376206.
[47] Frank S, Fürst C, Koschke L, Witt A, Makeschin F. Assessment of landscape aesthetics—validation of a landscape metrics-based assessment by visual estimation of the scenic beauty. Ecol Ind 2013;32:222–31. https://doi.org/10.1016/j.ecolind.2013.03.026.
[48] Gobster PH, Ribe RG, Palmer JF. Themes and trends in visual assessment research: Introduction to the Landscape and Urban Planning special collection on the visual assessment of landscapes. Landscape Urban Plann 2019;191:103635. https://doi.org/10.1016/j.landurbplan.2019.103635.
[49] Brown AL, Kang J, Gjestland T. Towards standardization in soundscape preference assessment. Appl Acoust 2011;72(6):387–92. https://doi.org/10.1016/j.apacoust.2011.01.001.
[50] Givoni B, Noguchi M, Saaroni H, Pochter O, Yaacov Y, Feller N, et al. Outdoor comfort research issues. Energy Build 2003;35(1):77–86. https://doi.org/10.1016/S0378-7788(02)00082-8.
[51] Sun K, De Coensel B, Filipan K, Aletta F, Van Renterghem T, De Pessemier T, et al. Classification of soundscapes of urban public open spaces. Landscape Urban Plann 2019;189:139–55. https://doi.org/10.1016/j.landurbplan.2019.04.016.
[52] Nilsson M, Botteldooren D, De Coensel B. Acoustic indicators of soundscape quality and noise annoyance in outdoor urban areas. Proceedings of the 19th International Congress on Acoustics, 2007.
[53] Axelsson Ã, Nilsson ME, Berglund B. The Swedish soundscape-quality protocol. J Acoust Soc Am 2012;131(4):3476. https://doi.org/10.1121/1.4709112.
[54] Berglund B, Nilsson M. On a tool for measuring soundscape quality in urban residential areas. Acta Acustica United with Acustica 2006;92:938–44.
[55] Nilsson M, Berglund B. Soundscape quality in suburban green areas and city parks. Acta Acustica United with Acustica 2006;92:903–11.
[56] Hong JY, Jeon JY. Influence of urban contexts on soundscape perceptions: A structural equation modeling approach. Landscape Urban Plann 2015;141:78–87. https://doi.org/10.1016/j.landurbplan.2015.05.004.
[57] Liu J, Kang J, Behm H. Birdsong as an element of the urban sound environment: A case study concerning the area of Warnemünde in Germany. Acta Acustica United with Acustica 2014;100(3):458–66. https://doi.org/10.3813/AAA.918726.
[58] Bell S (Ed.). *Elements of visual design in the landscape*. Oxfordshire: Routledge. (2013). https://doi.org/10.4324/9780203358146.
[59] Hehl-Lange S. Structural elements of the visual landscape and their ecological functions. Landscape Urban Plann 2001;54(1–4):107–15. https://doi.org/10.1016/S0169-2046(01)00129-3.
[60] Jakle JA, editor. The visual elements of landscape. Amherst: University of Massachusetts Press; 1987.
[61] Hong JY, Jeon JY. Relationship between spatiotemporal variability of soundscape and urban morphology in a multifunctional urban area: A case study in Seoul, Korea. Build Environ 2017;126:382–95. https://doi.org/10.1016/j.buildenv.2017.10.021.
[62] Liu J, Kang J, Luo T, Behm H. Landscape effects on soundscape experience in city parks. Sci Total Environ 2013;454:474–81. https://doi.org/10.1016/j.scitotenv.2013.03.038.
[63] Davies WJ, Adams MD, Bruce NS, Cain R, Carlyle A, Cusack P, et al. Perception of soundscapes: An interdisciplinary approach. Appl Acoust 2013;74(2):224–31. https://doi.org/10.1016/j.apacoust.2012.05.010.
[64] Kang J, Zhang M. Semantic differential analysis of the soundscape in urban open public spaces. Build Environ 2010;45(1):150–7. https://doi.org/10.1016/j.buildenv.2009.05.014.
[65] Kawai K, Kojima T, Hirate K, Yasuoka M. Personal evaluation structure of environmental sounds: experiments of subjective evaluation using subjects' own terms. J Sound Vib 2004;277(3):523–33. https://doi.org/10.1016/j.jsv.2004.03.013.
[66] Raimbault M, Lavandier C, Bérengier M. Ambient sound assessment of urban environments: field studies in two French cities. Appl Acoust 2003;64(12):1241–56. https://doi.org/10.1016/S0003-682X(03)00061-6.
[67] Axelsson Ö, Nilsson ME, Berglund B. A principal components model of soundscape perception. J Acoust Soc Am 2010;128(5):2836–46. https://doi.org/10.1121/1.3493436.
[68] Echelberger HE. The semantic differential in landscape research. *Proceedings of our national landscape: a conference on applied techniques for analysis and management of the visual resource* (General Technical Report PSW-GTR-35). Berkeley, CA. - U.S. Department of Agriculture, Forest Service, Pacific Southwest Research Station. Retrieved from: https://www.fs.fed.us/psw/publications/documents/psw_gtr035/psw_gtr035_12_echelberger.pdf (1979).
[69] Park BJ, Furuya K, Kasetani T, Takayama N, Kagawa T, Miyazaki Y. Relationship between psychological responses and physical environments in forest settings. Landscape Urban Plann 2011;102(1):24–32. https://doi.org/10.1016/j.landurbplan.2011.03.005.
[70] Song C, Ikei H, Kobayashi M, Miura T, Li Q, Kagawa T, et al. Effects of viewing forest landscape on middle-aged hypertensive men. Urban For Urban Greening 2017;21:247–52. https://doi.org/10.1016/j.ufug.2016.12.010.
[71] Yamashita S. Perception and evaluation of water in landscape: use of Photo-Projective Method to compare child and adult residents' perceptions of a Japanese river environment. Landscape Urban Plann 2002;62(1):3–17. https://doi.org/10.1016/S0169-2046(02)00093-2.
[72] Jeon JY, Hong JY. Classification of urban park soundscapes through perceptions of the acoustical environments. Landscape Urban Plann 2015;141:100–11. https://doi.org/10.1016/j.landurbplan.2015.05.005.
[73] Joo SH. Study on the selection and the application of landscape adjectives for the urban landscape analysis. J Korean Inst Landscape Architect 2003.
[74] Oh K. A perceptual evaluation of computer-based landscape simulations. Landscape Urban Plann 1994;28(93):201–16. https://doi.org/10.1016/0169-2046(94)90008-6.

[75] Shibata T, Kato T. Image retrieval system for street landscape images using adjectives. Syst Comput Japan 2001;32(6):62–71. https://doi.org/10.1002/scj.v32:610.1002/scj.1035.
[76] Aletta F, Guattari C, Evangelisti L, Asdrubali F, Oberman T, Kang J. Exploring the compatibility of "Method A" and "Method B" data collection protocols reported in the ISO/TS 12913–2: 2018 for urban soundscape via a soundwalk. Appl Acoust 2019;155:190–203. https://doi.org/10.1016/j.apacoust.2019.05.024.
[77] Myers L, Sirois MJ. Spearman correlation coefficients, differences between. In: Kotz S, Read CB, Balakrishnan N, Vidakovic B, Johnson NL, editors. Encyclopedia of Statistical Sciences. Hoboken: John Wiley & Sons; 2004. https://doi.org/10.1002/0471667196.ess5050.
[78] Var I. Multivariate data analysis. Vectors 1998;8(2):125–36.
[79] Jeon JY, Jo HI, Kim SM, Yang HS. Subjective and objective evaluation of water-supply and drainage noises in apartment buildings by using a head-mounted display. Appl Acoust 2019;148:289–99. https://doi.org/10.1016/j.apacoust.2018.12.037.
[80] Baños RM, Botella C, Alcañiz M, Liaño V, Guerrero B, Rey B. Immersion and emotion: their impact on the sense of presence. Cyberpsychol Behav Soc Netw 2004;7(6):734–41. https://doi.org/10.1089/cpb.2004.7.734.
[81] Brooks J, Lodge R, White D. Comparison of a head-mounted display and flat screen display during a micro-UAV target detection task. Proceedings of the Human Factors and Ergonomics Society Annual Meeting 2017;61(1):1514–8. https://doi.org/10.1177/1541931213601863.
[82] Azevedo RGDA Birkbeck N, De Simone F, Janatra I, Adsumilli B, Frossard P. Visual distortions in 360-degree videos. IEEE: (2019) 1901.01848. https://doi.org/10.1109/TCSVT.2019.2927344.
[83] Monteiro D, Liang H-N, Xu W, Brucker M, Nanjappan V, Yue Y. Evaluating enjoyment, presence, and emulator sickness in VR games based on first- and third-person viewing perspectives. Comput Anim Virtual Worlds 2018;29(3-4):e1830. https://doi.org/10.1002/cav.1830.
[84] Bowman DA, McMahan RP. Virtual reality: How much immersion is enough? IEEE Comput 2007;40(7):36–43. https://doi.org/10.1109/MC.2007.257.
[85] Seibert J, Shafer DM. Control mapping in virtual reality: effects on spatial presence and controller naturalness. Virtual Reality 2018;22(1):79–88. https://doi.org/10.1007/s10055-017-0316-1.
[86] Hong JY, Jeon JY. Designing sound and visual components for enhancement of urban soundscapes. J Acoust Soc Am 2013;134(3):2026–36. https://doi.org/10.1121/1.4817924.
[87] Ren X, Kang J, Zhu P, Wang S. Effects of soundscape on rural landscape evaluations. Environ Impact Assess Rev 2018;70:45–56. https://doi.org/10.1016/j.eiar.2018.03.003.
[88] Hou J, Nam Y, Peng W, Lee KM. Effects of screen size, viewing angle, and players' immersion tendencies on game experience. Comput Hum Behav 2012;28(2):617–23. https://doi.org/10.1016/j.chb.2011.11.007.
[89] Gorisse G, Christmann O, Amato EA, Richir S. First-and third-person perspectives in immersive virtual environments: Presence and performance analysis of embodied users. Front Robot AI 2017;4:33. https://doi.org/10.3389/frobt.2017.00033.
[90] Jo HI, Jeon JY. Downstairs resident classification characteristics for upstairs walking vibration noise in an apartment building under virtual reality environment. Build Environ 2019;150:21–32. https://doi.org/10.1016/j.buildenv.2018.12.05.
[91] Erfanian M, Mitchell A, Aletta F, Kang J. Psychological well-being and demographic factors can mediate soundscape pleasantness and eventfulness: A large sample study. J Environ Psychol 2021;77:101660. https://doi.org/10.1016/j.jenvp.2021.101660.
[92] Ragan ED, Bowman DA, Kopper R, Stinson C, Scerbo S, McMahan RP. Effects of field of view and visual complexity on virtual reality training effectiveness for a visual scanning task. IEEE Trans Visual Comput Graphics 2015;21(7):794–807. https://doi.org/10.1109/TVCG.2015.2403312.
[93] Witmer BG, Singer MJ. Measuring presence in virtual environments: A presence questionnaire. Presence 1998;7(3):225–40. https://doi.org/10.1162/105474698565686.